

Asutosh Panda

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Senior Software Engineer with 7+ years building **scalable**, high-scale enterprise backend systems — specialising in **distributed systems, microservices, and database engineering** across SQL and NoSQL stores. SME for critical production services, with a strong track record in **system performance, observability, and AI-native engineering**. Experienced in technical leadership, incident ownership, and engineering mentorship.

EXPERIENCE

[Aptos India Pvt. Ltd.](#), Bengaluru — Sr. Software Engineer (JAN 2023 – PRESENT)

Tech: React.js, React Native, TypeScript, Node.js, Java, PostgreSQL, MySQL, Couchbase, CrateDB, RabbitMQ, REST APIs

- **Subject Matter Expert:** Designated SME for Mobile POS, Store Accounting, and Promotion & Pricing Engine — owning architectural decisions, maintainability, and reliability roadmaps for multiple high-throughput production services.
- **Database re-engineering:** Introduced generated columns and index strategies to eliminate costly JSON-blob **query** extraction under peak load — achieving **~80% reduction in API latency** and eliminating timeout spikes at peak store hours.
- **Distributed & event-driven systems:** Built RabbitMQ-based async workflows with retry logic, dead-letter queues, and fault tolerance — ensuring reliable event distribution across enterprise microservices at scale.
- **Production incident response:** Led Tier-3 escalation as part of a rotating on-call support team — resolving critical hotfixes within **1–4 business days** and maintaining an average resolution time of **~3.5 business days** across all severity levels.
- **Observability & reliability:** Implemented distributed tracing and structured logging across microservices — eliminating recurring 500-series errors in store lifecycle flows and reducing MTTD for production incidents.
- **Promotion & Pricing Engine:** Refactored hierarchical search with pagination and caching, transforming multi-second SKU lookups into **near-instantaneous queries** across millions of records and guaranteeing **zero-downtime campaign execution** during high-traffic retail periods.
- **POS transaction state machine:** Redesigned React Native POS lifecycle to eliminate race conditions and async collisions — resolving critical checkout crashes across enterprise retail deployments.
- **AI-driven productivity:** Built internal AI agents to automate legacy component extraction and test generation — reducing migration effort by **~40%** and establishing a repeatable pattern that cut regression rates across the team.
- **Enterprise platform delivery:** Architected and delivered the Promotion Management Tool — a large-scale React platform for rule-driven in-store promotions, driving multi-client adoption and strong stakeholder feedback.
- **CI/CD stability:** Stabilised test suites and pipelines, reducing failure rates from **~40% to ~5%** — improving release reliability and deployment confidence.
- **Engineering mentorship:** Mentor and manage junior engineers through 1:1s, code reviews, and coaching on system design — improving onboarding efficiency and raising the overall engineering bar.
- **Technical documentation:** Authored 20+ design docs, ADRs, and release guides — creating runbooks and SOPs that reduced cross-team friction and accelerated onboarding globally.
- Recognised with the **Remarkable Colleague Award** and **Excellence Award** for technical ownership, delivery excellence, and incident response leadership.

[Bazaarvoice](#), Bengaluru — Software Engineer (MAR 2021 - JAN 2023)

Tech: Java (Dropwizard), React.js, Node.js, Express, TypeScript, Google Apigee, PostgreSQL, Docker, GCP

Key Initiatives: *Privacy Service Modernization | Video Submission for Ratings & Reviews | Key Management Platform*

- **Microservices migration:** Led the Privacy Service move from monolith to DDD-based microservices, improving deployment speed and modularity by **~70%**.

- **Intelligent data archiving:** Added data archiving to stop unchecked database growth, cutting storage costs by **70%** and reducing sale-period outages to **near zero**.
- **Zero-downtime migration:** Delivered the migration with zero downtime, **two weeks ahead of deadline**, by phasing the rollout across teams; earned the **Quarterly Spot Award**.
- **Service consolidation:** Merged three Key Management Services into one platform, cutting infrastructure costs by **50%** and maintenance effort by **~40%**.
- **API Gateway:** Owned and optimised **150+** Apigee proxy endpoints, improving API reliability and routing efficiency.
- **Observability:** Added structured logging, tracing, and alerting on GCP, speeding up incident response.
- **Video Submission:** Designed and shipped the Video Submission feature for Ratings & Reviews independently, on time.
- **Frontend development:** Built React.js applications for the Privacy and Key Management services.
- Recognised with the **Tech Excellence Award 2022** for high-impact architecture and mentoring.

[Skylark Drones](#), Bengaluru — Full Stack Developer (MAY 2019 - FEB 2021)

Tech: Vue.js, D3.js, Django, Flask, PostgreSQL, PostGIS, MongoDB, CesiumJS, GitLab CI/CD

Key Projects: *Spectra* | *Covid-19 Surveillance Dashboard* | *Drone Mission Control*

- **Full-stack platform ownership:** Built end-to-end web platforms for drone analytics, inspection, and real-time surveillance — owning frontend, backend, and production deployments across multiple high-impact products.
- **Geospatial data visualisation:** Built data-intensive frontends using Vue.js and D3.js — implementing map annotations, clustering, filters, timelines, and synchronised video playback for mission-critical drone operations.
- **High-volume image pipeline:** Engineered an upload and processing pipeline supporting **7,000+ images per survey** with batching, retry mechanisms, and backend integration.
- **Data tools & operational dashboards:** Built role-based data management tools for accessing and downloading large drone mission datasets; created Gantt-style planning dashboards for tracking pilots, drones, and projects.
- **Frontend performance:** Reduced bundle size by **~15%** via lazy loading and dependency pruning, and cut network usage by **~30%** through API optimisations.
- **Backend engineering:** Contributed to analytics and processing services using Django, Flask, PostgreSQL, PostGIS, and MongoDB — handling geospatial queries and system integrations across drone mission workflows.
- **CI/CD pipelines:** Set up and maintained GitLab CI/CD pipelines for reliable, repeatable deployments to staging and production for mission-critical systems.

SKILLS

Backend: Node.js, Express.js, Flask, Django, Java (Dropwizard), RESTful API design, Microservices, Event-driven architecture (RabbitMQ)

Frontend: HTML, CSS, JavaScript, Typescript, VueJs, ReactJS

Databases & ETL: MySQL, PostgreSQL, MongoDB, Couchbase, CrateDB, RabbitMQ (async workflows, retry, DLQ), Apache SOLR

Cloud & DevOps: GCP, AWS, Docker, GitLab CI/CD, CI pipeline stability

Observability: Structured logging, distributed tracing, alerting, monitoring baselines, incident response

AI Tooling: Cursor AI, GitHub Copilot, ChatGPT/Claude — daily use for coding, debugging, code review, and automation

Other: React Native, Apollo/GraphQL, D3.js, RxJS, Redux, TypeScript, Rust, Git

EDUCATION

College of Engineering & Technology, Bhubaneswar — B.Tech CSE | 2015–2019 | CGPA 8.78